KO LIH HAN

Software Engineer | Applied AI & LLM Systems

Malaysia | +60 17-424 6282 | lihhanko526@gmail.com | github.com/kolihhan | LinkedIn

Software engineer with **4 years of part-time production experience** owning backend systems from build to deployment. Combines production engineering with hands-on LLM post-training, agent runtimes, retrieval, and evaluation.

EXPERIENCE

Software Developer (Part-time)

Kahap Inc

Aug 2022 – Aug 2026

- Owned the **0-to-1 build and production rollout** of a feedback platform, centralizing submissions, discussion, notifications, and admin workflows into one operational system.
- Kept a Laravel/Vue.js career platform running across queue workers, scheduled jobs, SSL renewals, database/server migrations, and Linux hosting.
- Integrated GPT-SoVITS and automated pronunciation/text checks into production features, connecting external AI services to existing backend workflows.

Full-Stack Developer Intern

Pac Com Solution

Jun 2021 – Jan 2022

- Shipped and debugged Laravel/Vue.js features in a production POS codebase across backend and frontend paths.

PROJECTS

Small-Agent QLoRA

github.com/kolihhan/small-agent-qlora

Python, Qwen3.5-9B, QLoRA / PEFT, Transformers, llama.cpp

- Post-trained a Qwen3.5-9B tool agent with QLoRA, lifting exact task success from **34.5% to 62.0%** on a frozen 200-task closed-loop evaluation.
- Built the agent runtime around bounded tool execution, duplicate-call blocking, timeouts, and persisted traces so failures could be reproduced and analyzed.
- Validated beyond the controlled benchmark on GAIA100, **doubling accuracy from 9% to 18%** and tracing the main remaining failure to over-search rather than tool errors.

Semantic Text-to-SQL

github.com/kolihhan/semantic-text2sql-agent

Python, FastAPI, LangGraph, SQLite, Qwen3.5-9B

- Designed and built a local NL-to-SQL service with schema grounding, deterministic validation, bounded repair, and read-only execution to reject invalid or destructive queries before they reach the database.
- Improved execution success from **75% to 90%** and execution-match accuracy from **33% to 39%** on 100 frozen BIRD DEV queries, recovering 6 wrong answers with 0 regressions.
- Unified FastAPI, CLI, and Streamlit behind one service layer so generation, verification, execution, and evaluation reused the same backend logic.

RESEARCH

Context-Aware Workplace Language Understanding

M.S. Thesis, National Tsing Hua University

NLP, retrieval, role/context graphs

- Curated and labeled **20,455 workplace utterances across 7 boards**, capturing speaker/listener state, emotion, and communicative purpose for implicit-meaning analysis.
- Designed role- and context-aware graph retrieval/fusion, improving target-only macro-F1 from **0.625 to 0.689** on 401 test utterances.

SKILLS

AI / LLM: Hugging Face Transformers, PEFT / QLoRA, LangGraph, llama.cpp, RAG, tool calling, LLM evaluation

Backend: Python, FastAPI, PHP, Laravel, SQL, REST APIs

Data / Infrastructure: MySQL, SQLite, Linux, Git, GitHub Actions, pytest

Frontend: JavaScript, Vue.js, React

Languages: Chinese (Native/Bilingual), English (Fluent), Malay (Intermediate)

EDUCATION

National Tsing Hua University

Master of Science, Information Systems & Applications | GPA: 4.09/4.3 | Jun 2026

Chung Hua University

Bachelor of Science, Computer Science & Information Engineering | GPA: 3.9/4.0 | Jun 2024